

Acoustic Interaction Force Between Two Identical Compressible Particles Trapped in an Acoustofluidic Nodal Plane

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Abstract

We derive a closed-form expression for the acoustic interaction force between two identical compressible particles trapped in an acoustofluidic nodal plane. Working in the long-wavelength scattering regime for an ideal fluid, we solve the coupled scattering problem within a fifth-order multipole expansion. The resulting force law extends the Silva–Bruus theory through explicit finite-separation corrections, while recovering it in the weak-density-contrast and large-separation limits. In the close-range regime, the interaction force retains the characteristic inverse-fourth-power dependence on interparticle distance, with a prefactor renormalized by repeated rescattering. Comparison with previous numerical calculations and experiments shows excellent agreement with ideal-fluid numerics in the weak-density-contrast regime, but substantially larger deviations from several experiments, indicating that viscosity, acoustic streaming, and hydrodynamic interactions remain important for quantitative prediction.

Keywords: Acoustic secondary radiation force, Acoustic microstreaming flow, Acoustic trapping

1 Introduction

Acoustic manipulation of micro- and nano-scale particles has become a versatile, label-free, and biocompatible method for particle sorting, concentration, rotation, and assembly in microfluidic systems [1, 2]. This area is broadly known as *acoustofluidics*. Its underlying physics is governed by two distinct but coupled mechanisms: acoustic

radiation forces, arising from the time-averaged momentum flux of scattered sound waves [3], and acoustic microstreaming, namely the steady secondary flow generated by viscous dissipation of oscillatory acoustic motion in boundary layers and in the bulk fluid [4].

A large class of acoustofluidic devices operates with ultrasonic standing waves, in which subwavelength particles are driven toward pressure nodes or antinodes according to their density and compressibility contrasts with the suspending fluid [5, 6]. Rigid particles are typically trapped at pressure nodes. Once particles accumulate at these equilibrium locations, their subsequent collective dynamics are no longer governed solely by the primary radiation force. Instead, secondary acoustic interaction forces and streaming-induced hydrodynamic coupling become important, controlling aggregation, ordering, and the formation of particle assemblies and colloidal crystals [7].

The theory of acoustic radiation force on isolated particles is well established. King [8] provided the classical analysis for a rigid sphere in an inviscid fluid, and Gor'kov [9] later showed that, in the long-wavelength limit, the force induced by a stationary acoustic field can be written as the gradient of an interaction potential. Extensions to viscous fluids were developed by Doinikov [10], who explicitly retained second-order viscous effects associated with streaming, and by Settnes & Bruus [11], who generalized Gor'kov's result to a viscous host fluid while excluding the direct force contribution of second-order microstreaming.

The theoretical description of *secondary* acoustic radiation forces has a comparably long history. König [12] already interpreted acoustic attraction and repulsion in terms of the mutual disturbance of oscillatory flows around nearby bodies. Embleton [13] later reformulated the problem in a clearer scattering framework for two spheres in a plane sound field. Subsequent studies established that the interaction force depends sensitively on particle separation, orientation, material contrast, and rescattering. In particular, Zheng [14] treated two fluid spheres aligned with the propagation direction of traveling and standing plane waves, Weiser *et al.* [15] related interaction forces in standing waves to red-blood-cell spacing, and Doinikov & Zavtrak [16] showed that close-range bubble–solid interactions may differ qualitatively from their far-field asymptotics.

More general formulations were developed later. Doinikov [17] derived a compressible-fluid multiple-scattering theory valid for arbitrary particle sizes and separations. In the Rayleigh limit, Silva & Bruus [18] obtained a pair-interaction theory based on monopole and dipole scattering, which has become a standard analytical reference for small-particle interaction forces. Lopes *et al.* [19] extended the multiple-scattering formulation to Mie-sized compressible spheres, including both forces and torques. For standing-wave configurations, Sepehrirahnama *et al.* [20] computed interaction forces between rigid particles, while Sepehrirahnama & Lim [21] introduced a generalized multibody scattering formulation suitable for close-range interactions. In parallel, several studies emphasized the role of viscous effects: Sepehrirahnama *et al.* [22] showed numerically that microstreaming can substantially enhance interparticle forces, and Pavlic *et al.* [23] demonstrated that it may even reverse the sign

of the interaction force for elastic subwavelength spheres in a standing wave. Many-particle dynamics including radiation forces, hydrodynamic resistance, and contact interactions were further examined by Baasch *et al.* [24].

Experimental measurements of the secondary acoustic radiation force for Rayleigh-scale elastic particles have also become available [25–29]. These studies broadly confirm the qualitative trends predicted by theory, but quantitative discrepancies remain common. In particular, interaction forces measured at close range are often larger than those predicted by ideal-fluid analytical models, whereas improved agreement has been reported when higher-order rescattering is incorporated. This suggests that the respective roles of finite-separation multiple scattering, viscous effects, and streaming-induced hydrodynamic coupling still need to be more clearly disentangled.

In this work, we derive a closed-form expression for the acoustic interaction force between two identical compressible particles trapped in the nodal plane of a standing-wave field. The analysis is carried out in the Rayleigh regime for an ideal fluid, so that the contribution of acoustic multiple scattering can be isolated from viscous microstreaming effects. By exploiting the symmetry of the nodal-plane two-particle configuration, the coupled scattering problem is reduced to a tractable multipole system, from which an explicit analytical interaction-force formula is obtained. The resulting theory recovers the Silva–Bruus acoustic interaction force [18] in the appropriate asymptotic limits, while extending it through finite-separation corrections associated with repeated mutual rescattering. This allows direct comparison with earlier analytical theories [15], higher-order numerical calculations [21], and experimental measurements reported in the literature. In this way, the present result helps delineate the range of validity of ideal-fluid pair-interaction models and clarifies when additional physical effects, beyond ideal-fluid multiple scattering, are required to account for the observed interaction force.

2 Fluid dynamics model

The acoustic field is described in terms of the mass density $\rho(\mathbf{r}, t)$, pressure $p(\mathbf{r}, t)$, and fluid velocity $\mathbf{v}(\mathbf{r}, t)$ as a function of position vector $\mathbf{r}$ and time t . The fluid is at rest, which is characterized by a uniform ambient state with density ρ_0 , pressure p_0 , and adiabatic sound speed c_0 . The corresponding isentropic compressibility is $\kappa_0 = (\rho_0 c_0^2)^{-1}$. The acoustic perturbations are assumed to evolve under isentropic conditions in an ideal fluid, so that viscous and thermal effects are neglected.

The governing equations consist of mass conservation, Euler’s equation, and a linear equation of state. These are written as

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (1a)$$

$$\rho (\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v}) = -\nabla p, \quad (1b)$$

$$p = c_0^2 \rho. \quad (1c)$$

In the present analysis, the low-amplitude wave assumption is taken, where the wave characteristic velocity v_0 is much smaller than the speed of sound in the fluid.

The fluid dynamics fields can be conveniently expanded as

$$\rho(\mathbf{r}, t) = \rho_0 + \rho^{(1)}(\mathbf{r})e^{-i\omega t}, \quad (2a)$$

$$p(\mathbf{r}, t) = p_0 + p^{(1)}(\mathbf{r})e^{-i\omega t}, \quad (2b)$$

$$\mathbf{v}(\mathbf{r}, t) = \mathbf{0} + \mathbf{v}^{(1)}(\mathbf{r})e^{-i\omega t}, \quad (2c)$$

where ω is the angular frequency of the acoustic wave. Substituting these equations into Eqs. (1a)–(1c) and retaining only first-order terms, we obtain the linear acoustic equations

$$-i\omega\rho^{(1)} + \rho_0\nabla \cdot \mathbf{v}^{(1)} = 0, \quad (3a)$$

$$-i\omega\rho_0\mathbf{v}^{(1)} = -\nabla p^{(1)}, \quad (3b)$$

$$p^{(1)} = c_0^2\rho^{(1)}. \quad (3c)$$

Eliminating $\rho^{(1)}$ and $\mathbf{v}^{(1)}$ from Eq. (3), one finds

$$\nabla \cdot \mathbf{v}^{(1)} = \frac{i\omega}{\rho_0 c_0^2} p^{(1)}, \quad \mathbf{v}^{(1)} = -\frac{i}{\rho_0 \omega} \nabla p^{(1)}. \quad (4)$$

Taking the divergence of the second relation and substituting the first one into the result, we arrive at

$$\nabla^2 p^{(1)} + k^2 p^{(1)} = 0, \quad (5)$$

where $k = \omega/c_0$ is the acoustic wavenumber. Therefore, in an ideal fluid, the first-order pressure field satisfies the Helmholtz equation.

3 Acoustic scattering theory

3.1 Linear single-particle scattering

We first consider the scattering response of a compressible spherical particle, labeled by i , with radius a_i , density ρ_i , speed of sound c_i , and compressibility $\kappa_i = (\rho_i c_i)^{-1}$. In the long-wavelength (Rayleigh) regime, $ka_i \ll 1$. We assume the spherical coordinate system with respect to the i th-particle as $\mathbf{r}_i = (r_i, \theta_i, \varphi_i)$. Moreover, it is common to write pressure and fluid velocity normalized as

$$\tilde{p} = \sqrt{\frac{\kappa_0}{4E_0}} p, \quad \tilde{\mathbf{v}} = \sqrt{\frac{\rho_0}{4E_0}} \mathbf{v} \quad (6)$$

where $E_0 = \rho_0 v_0^2/4$ is the acoustic characteristic energy density. Hereafter, we dropped the superscript ⁽¹⁾ which denotes linear approximation.

The incident (external) field admits the partial-wave expansion related to particle i reference frame,

$$\tilde{p}_i^{\text{ext}}(\mathbf{r}_i) = \sum_{nm} a_{nm}^{(i)} j_n(kr_i) Y_n^m(\Omega_i), \quad (7)$$

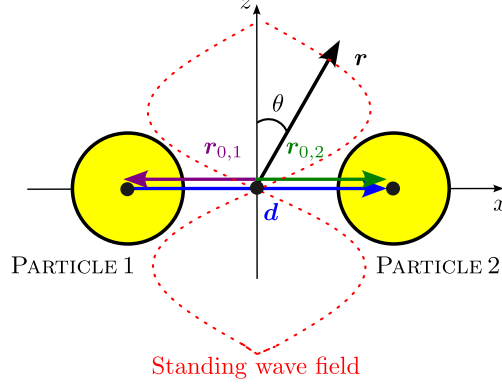


Fig. 1 Two-particle scattering configuration in an acoustic standing-wave field. The particles are located in the pressure nodal plane, perpendicular to the direction of wave propagation.

where $\sum_{nm} = \sum_{n=0}^{\infty} \sum_{m=-n}^n$, j_n is the spherical Bessel function, Y_n^m are spherical harmonics, $\Omega_i = (\theta_i, \varphi_i)$. The beam-shape coefficients a_{nm} are given by

$$a_{nm}^{(i)} = \frac{1}{j_n(ka_i)} \int \tilde{p}^{\text{ext}}(a_i, \Omega_i) Y_n^{m*}(\Omega_i) d\Omega_i. \quad (8)$$

For a single spherical particle i , the scattered pressure field is expanded as

$$\tilde{p}_i^{\text{sc}}(\mathbf{r}_i) = \sum_{nm} s_n^{(i)} a_{nm}^{(i)} h_n(kr_i) Y_n^m(\Omega_i), \quad (9)$$

h_n is the spherical Hankel function of the first kind. The scaled scattering coefficient $s_n^{(i)}$ for a suspended particle is given by [30]

$$s_n^{(i)} = -\det \begin{bmatrix} \frac{\rho_0 k_i}{\rho_i k} j_n(ka_i) & j_n(k_i a_i) \\ j_n'(ka_i) & j_n'(k_i a_i) \end{bmatrix} \det \begin{bmatrix} \frac{\rho_0 k_i}{\rho_i k} h_n(ka_i) & j_n(k_i a_i) \\ h_n'(ka_i) & j_n'(k_i a_i) \end{bmatrix}^{-1}, \quad (10)$$

where $k_i = \omega/c_i$ is the wavenumber inside the particle.

3.2 Linear multiparticle scattering

We consider the linear acoustic scattering problem for a system of N_p small spherical particles, described by a truncated multipole expansion. The particles, labeled by $i = 1, 2, \dots, N_p$, have radii a_i , densities ρ_i , and compressibilities κ_i , and their centers are initially located at $\mathbf{r}_{0,i}$. In Fig. 1, we depict the two-particle scattering problem with the interparticle vector distance denoted by $\mathbf{d}$. The particle placed symmetrically in the x axis, which coincides to a pressure nodal plane of a standing wave field.

Each particle is excited not only by the external field but also by the fields scattered by all other particles. Accordingly, the effective incident field on particle i is

$$\tilde{p}_i^{\text{in}}(\mathbf{r}_i) = \tilde{p}_i^{\text{ext}}(\mathbf{r}_i) + \sum_{j \neq i}^{N_p} \tilde{p}_j^{\text{sc}}(\mathbf{r}_i). \quad (11)$$

This coupling is essential even in the Rayleigh limit when the interparticle distance is comparable to the particle size.

To evaluate its contribution to the effective incident field on particle i , this scattered field is re-expanded about the coordinate system of particle i , yielding [31]

$$\tilde{p}_j^{\text{sc}}(\mathbf{r}_i) = \sum_{nm}^N b_{nm}^{(j \rightarrow i)} j_n(kr_i) Y_n^m(\Omega_i), \quad b_{nm}^{(j \rightarrow i)} = \sum_{n'm'}^N S_{n'n}^{m'm}(\mathbf{r}_{ij}) s_{n'm'}^{(j)}, \quad (12)$$

where $\mathbf{r}_{ij} = \mathbf{r}_{0,i} - \mathbf{r}_{0,j}$ and $r_i < r_{ij}$. The scattered pressure $\tilde{p}_j^{\text{sc}}$ is truncated at the N th multipole order. The separation coefficient is given by

$$S_{n'n}^{m'm}(\mathbf{r}_{ij}) = 4\pi i^{n-n'} \sum_{l=0}^{\frac{n+n'-q_0}{2}} i^{q_0+2l} h_{q_0+2l}^{(1)}(kr_{ij}) Y_{q_0+2l}^{m-m'}(\Omega_{ij}) \times \mathcal{G}(n', m'; q_0 + 2l, m - m'; n, m), \quad (13)$$

where q_0 is the smallest integer such that $q_0 \geq \max(|n - n'|, |m - m'|)$ and $q_0 \bmod 2 = (n + n') \bmod 2$. The Gaunt coefficient is defined as

$$\mathcal{G}(n', m'; q, m - m'; n, m) = \int Y_{n'}^{m'}(\Omega) Y_q^{m-m'}(\Omega) Y_n^{-m}(\Omega) d\Omega. \quad (14)$$

The effective incident pressure to particle i is then

$$\tilde{p}_i^{\text{eff}}(\mathbf{r}_i) = \sum_{nm}^N \left[a_{nm}^{(i)} + \sum_{j \neq i}^{N_p} b_{nm}^{(j \rightarrow i)} \right] j_n(kr_i) Y_n^m(\Omega_i). \quad (15)$$

The coefficients $a_{nm}^{(i)}$ are the partial-wave expansion coefficients of the external field about the center of particle i . Hence, the coupled rescattering system obtained from the interfacial continuity conditions in the multisphere formulations can be written as [19, 21]

$$s_{nm}^{(i)} = s_n^{(i)} \left[a_{nm}^{(i)} + \sum_{j \neq i}^{N_p} \sum_{n'm'}^N S_{n'n}^{m'm}(\mathbf{r}_{ij}) s_{n'm'}^{(j)} \right]. \quad (16)$$

3.3 Multiple scattering in a standing plane wave

Consider a one-dimensional plane standing wave developed in an ideal fluid. The dimensionless first-order pressure field is taken as

$$\tilde{p}^{\text{ext}} = \cos[k(z + z_0)], \quad (17)$$

where z_0 is a spatial offset. Using the linear momentum relation in (4), the corresponding axial velocity is

$$\tilde{v}_z^{\text{ext}} \approx -\frac{i}{k} \partial_z \tilde{p}^{\text{ext}} = i \sin[k(z + z_0)]. \quad (18)$$

The primary acoustic radiation force drives suspended particles toward either pressure nodes or pressure antinodes, depending on the sign of the acoustophoretic factor [5],

$$\Phi^{(i)} = \frac{1}{3} f_0^{(i)} + \frac{1}{2} f_1^{(i)}. \quad (19)$$

Specifically, particle i is trapped in a pressure nodal plane when $\Phi^{(i)} > 0$, and in a pressure antinodal plane when $\Phi^{(i)} < 0$. Here, $f_0^{(i)}$ and $f_1^{(i)}$ are the monopole and dipole scattering factors,

$$f_0^{(i)} = 1 - \tilde{\kappa}_i, \quad f_1^{(i)} = 2 \frac{\tilde{\rho}_i - 1}{2\tilde{\rho}_i + 1}, \quad (20)$$

where $\tilde{\kappa}_i = \kappa_i / \kappa_0$ and $\tilde{\rho}_i = \rho_i / \rho_0$.

4 Acoustic interaction force

The total time-averaged acoustic radiation force acting on particle i is given by [11]

$$\mathbf{F}_i^{\text{rad}} = -\pi a_i^3 \left[\frac{2\kappa_0}{3} \text{Re} \left\{ f_0^{(i)*} p_i^{\text{in}*} \nabla_i p_i^{\text{in}} \right\} - \rho_0 \text{Re} \left\{ f_1^{(i)*} \mathbf{v}_i^{\text{in}*} \cdot \nabla_i \mathbf{v}_i^{\text{in}} \right\} \right]_{\mathbf{r}_i=\mathbf{0}}, \quad (21)$$

where ∇_i is the gradient operator with respect to particle i . The first term in this equation represents the monopole contribution associated with the compressibility contrast, while the second term captures the dipole contribution, which accounts for both the density contrast and viscous boundary-layer effects. The incident fluid velocity $\mathbf{v}^{\text{in}}$ is computed using Eq. (4).

Substituting Eq. (11) into (21), we obtain the acoustic interaction force $\mathbf{F}_i^{\text{int}}$ over particle i as

$$\begin{aligned} \mathbf{F}_i^{\text{int}} = -\pi a_i^3 \sum_{j \neq i}^N \text{Re} & \left[\frac{2\kappa_0}{3} f_0^{(i)*} (p_i^{\text{ext}*} \nabla p_j^{\text{sc}} + p_j^{\text{sc}*} \nabla p_i^{\text{ext}}) \right. \\ & \left. - \rho_0 f_1^{(i)*} (\mathbf{v}_i^{\text{ext}*} \cdot \nabla \mathbf{v}_j^{\text{sc}} + \mathbf{v}_j^{\text{sc}*} \cdot \nabla \mathbf{v}_i^{\text{ext}}) \right]_{\mathbf{r}_i=\mathbf{0}}. \end{aligned} \quad (22)$$

4.1 Particles trapped in a nodal plane

Assume that the suspended particles have a positive acoustophoretic factor, $\Phi^{(i)} > 0$. In this case, the primary acoustic radiation force drives them toward a pressure nodal plane. For convenience, we choose the xy plane, i.e. $z = 0$, to coincide with such a node. From Eq. (17), this requires $z_0 = -\frac{\pi}{2k}$. Accordingly, the external standing-wave field becomes $\tilde{p}^{\text{ext}} = \sin(kz)$. At the trapping plane $z = 0$, the external pressure vanishes, $\tilde{p}^{\text{ext}}(\mathbf{r}_{0,i}) = 0$, whereas its gradient is purely axial, $\nabla \tilde{p}^{\text{ext}}(\mathbf{r}_{0,i}) = k \mathbf{e}_z$. From Eq. (8), we find the beam-shape coefficients as

$$a_{n,0}^{(i)} = \begin{cases} 0, & n \text{ even}, \\ (-1)^{(n-1)/2} \sqrt{4\pi(2n+1)}, & n \text{ odd}, \end{cases} \quad (23)$$

with $a_{nm}^{(i)} = 0$ for all $m \neq 0$.

We now specialize the scattering-coefficient system in Eq. (16) using Eq. (23). It follows that

$$s_{nm}^{(i)} = s_n^{(i)} \left[\delta_{m,0} a_{n,0}^{(i)} + \sum_{j \neq i}^{N_p} b_{nm}^{(j \rightarrow i)} \right], \quad (24)$$

where δ_{ab} denotes the Kronecker delta. Therefore, in a pressure nodal plane the external standing wave directly forces only the axisymmetric multipoles $m = 0$, whereas components with $m \neq 0$ arise solely through multiple scattering among the particles.

We restrict the analysis to the case of two identical particles ($N_p = 2$), each of radius a . Owing to particle identity, the single-particle scattering coefficients are independent of i , so that $s_n^{(i)} = s_n$. The particles are initially located at $\mathbf{r}_{0,1} = (-d/2, 0, 0)$, $\mathbf{r}_{0,2} = (d/2, 0, 0)$, and $\mathbf{r}_{12} = -\mathbf{d} = -d \mathbf{e}_x$. This configuration possesses several symmetries that strongly reduce the number of independent scattering coefficients. Under the symmetry constraints analyzed in Appendix B, the present two-particle problem only admits

$$s_{n,-m}^{(i)} = s_{nm}^{(i)}, \quad s_{nm}^{(2)} = s_{nm}^{(1)}, \quad n \text{ odd}, \quad m = 0, 2, 4, \dots, n-1. \quad (25)$$

Hence, all independent unknowns may be taken from a single particle only, $s_{nm} = s_{nm}^{(1)} = s_{nm}^{(2)}$. Under these constraints Eq. (24) reduces to

$$s_{nm} = s_n \left[\delta_{m,0} a_{n,0} + \sum_{\substack{n'=1 \\ n' \text{ odd}}}^N \sum_{\substack{m'=0 \\ m' \text{ even}}}^{n'-1} \left(S_{n'n}^{m'm}(-\mathbf{d}) + (1 - \delta_{m',0}) S_{n'n}^{-m',m}(-\mathbf{d}) \right) s_{n'm'} \right], \quad (26)$$

with $n = 1, 3, 5, \dots, N$ and $m = 0, 2, 4, \dots, n-1$. The separation coefficients follow directly from Eq. (13), although their explicit evaluation is laborious. The number $\mathcal{N}(N)$ of unknown scattering coefficients in this linear system is given in Eq. (B8).

The interaction radiation force formula for the two-particle configuration is now derived. Replacing Eq. (12) into (22), we arrive at

$$\mathbf{F}^{\text{int}} = 4\pi a^3 E_0 \text{Re} \left[i f_1^* (\mathbf{e}_z \cdot \nabla_1) \tilde{\mathbf{v}}_2^{\text{sc}} \right]_{\mathbf{r}_1=0}, \quad (27)$$

with $f_1 = f_1^{(1)}$ and $\mathbf{F}^{\text{int}} = \mathbf{F}_1^{\text{int}}$. Now using (12) into this equation, we find the interaction force in the two-particle problem is purely directed along the line of centers,

$$\mathbf{F}^{\text{int}} = -\frac{2\sqrt{30}\pi}{15k^2} (ka)^3 E_0 \text{Re} \left[f_1^* b_{21}^{(2 \rightarrow 1)} \right] \mathbf{e}_x, \quad (28)$$

where $\mathbf{e}_x = \mathbf{e}_{x_1}$. The coefficient $b_{21}^{(2 \rightarrow 1)}$ is given by

$$b_{21}^{(2 \rightarrow 1)} = \sum_{\substack{n=1 \\ n \text{ odd}}}^N \sum_{\substack{m=0 \\ m \text{ even}}}^{n-1} \left[S_{n,2}^{m,1}(-\mathbf{d}) + (1 - \delta_{m,0}) S_{n,2}^{-m,1}(-\mathbf{d}) \right] s_{nm}. \quad (29)$$

The interaction force is ultimately determined by the reduced set of scattering coefficients s_{nm} . The central difficulty of the problem is therefore the solution of the coupled linear system in Eq. (26), from which all coefficients entering $b_{21}^{(2 \rightarrow 1)}$ must be obtained.

Although the symmetry reduction derived in Appendix B considerably decreases the number of independent unknowns, the resulting system remains nontrivial because the coefficients are coupled through the separation matrix and depend sensitively on the interparticle distance. In particular, close particle spacing enhances the importance of multiple scattering, so that a sufficiently high truncation order must be retained in the multipole expansion. Following Sepehrihahnama et al. [21], we adopt a minimum truncation order of $N = 5$, which was shown to provide agreement with experimental measurements, involving polystyrene microparticles in close-range, at the level of approximately 5%.

In the present two-particle nodal-plane problem, Eq. (B8) shows that the reduced system at $N = 5$ contains $\mathcal{N}(5) = 6$ independent coefficients, namely $s_{10}, s_{30}, s_{32}, s_{50}, s_{52}, s_{54}$. These correspond to the three nonvanishing multipole modes $n = 1, 3, 5$, allowed by the symmetry constraints of the problem. In Appendix D, the coefficient $b_{12}^{(2 \rightarrow 1)}$ is calculated to the order $\mathcal{O}(\epsilon^2)$, so that substituting Eq. (D20) into Eq. (28) leads us to

$$\mathbf{F}^{\text{int}} = 4\pi a^2 f_1^2 E_0 \left(\frac{a}{d} \right)^4 \left[\frac{3A_0}{D_0} (\cos(kd) + kd \sin(kd)) + \frac{A_2}{5D_0} (kd)^2 \cos(kd) \right] \mathbf{e}_x, \quad (30a)$$

$$A_0 = 4(77 - 25f_1 + 2f_1^2) + 90f_1(-11 + 2f_1)\chi^7 + 5250(-7 + f_1)f_1\chi^{11} + 33075f_1^2\chi^{18}, \quad (30b)$$

$$A_2 = -20(77 - 25f_1 + 2f_1^2) - 162f_1(-11 + 2f_1)\chi^7 - 24250(-7 + f_1)f_1\chi^{11}$$

$$+ 20025f_1^2\chi^{18}, \quad (30c)$$

$$\begin{aligned} D_0 = & -616 + 4f_1 (50 + 77\chi^3 + 1485\chi^7 + 22050\chi^{11}) \\ & + f_1^3\chi^3 (8 + 324\chi^7 + 5850\chi^{11} + 73575\chi^{18}) \\ & - 2f_1^2 [8 + 50\chi^3 + 9\chi^7 (60 + \chi^3(99 + 700\chi + 2275\chi^4 + 17850\chi^8))], \end{aligned} \quad (30d)$$

with $\chi = a/d$.

The magnitude of the acoustic interaction force is proportional to the particle surface area $4\pi a^2$, the acoustic energy density E_0 and to f_1^2 , indicating that the interaction is controlled by dipole-type scattering from both particles. The prefactor $(a/d)^4$ makes explicit the strong dependence on the interparticle distance. The trigonometric factors $\cos(kd)$ and $\sin(kd)$ describe the phase accumulated between the particles and therefore modulate both the magnitude and sign of the force. Finally, the coefficients A_0 , A_2 , and D_0 , which depend on f_1 and on the ratio a/d , collect the finite-separation corrections arising from the reduced $N = 5$ multiple-scattering solution.

5 Interaction force asymptotic analysis

In the nodal-plane configuration, the physically admissible interparticle distance is restricted by the non-overlap condition, $2a \leq d < \infty$, or, equivalently, $0 < a/d \leq 1/2$. Within this interval, three limiting regimes are especially relevant for interpreting Eq. (30a). The first is the weak-density-contrast regime, in which the particle density approaches that of the surrounding fluid, so that $f_1 \rightarrow 0$. This limit is frequently encountered in acoustofluidic applications involving biological cells and some polymer microparticles, for which the dipolar scattering response is weak. The second is the close-range regime, corresponding to $a/d \rightarrow 1/2$, where the particles approach near contact while remaining acoustically small compared with the wavelength. The third is the long-range regime, characterized by $a/d \rightarrow 0$, where the particles are geometrically well separated and repeated mutual rescattering becomes progressively weaker. A particularly important subregime of the latter is $kd \gg 1$, in which the interparticle spacing is also large compared with the acoustic length scale k^{-1} .

5.1 Weak-density-contrast limit

In the limit of weak density contrast, $f_1 \ll 1$, the interaction force in Eq. (30a) preserves the same oscillatory dependence on kd , whereas the finite-separation amplitude ratios become perturbative in f_1 . Expanding A_0/D_0 and A_2/D_0 about $f_1 = 0$, we obtain

$$\frac{A_0}{D_0} = -\frac{1}{2} - \frac{1}{308} (77\chi^3 + 990\chi^7 + 3675\chi^{11}) f_1 + O(f_1^2), \quad (31a)$$

$$\frac{A_2}{D_0} = \frac{5}{2} + \frac{1}{308} (385\chi^3 + 6534\chi^7 + 25375\chi^{11}) f_1 + O(f_1^2). \quad (31b)$$

Hence, to leading order in f_1 , Eq. (30a) reduces to

$$\mathbf{F}^{\text{int}}|_{f_1 \rightarrow 0} = 4\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \left[-\frac{3}{2} [\cos(kd) + kd \sin(kd)] + \frac{1}{2} (kd)^2 \cos(kd) \right] \mathbf{e}_x + O(f_1^3). \quad (32)$$

Therefore, the as the dipolar scattering response becomes weak, the finite-separation corrections become independent on both a/d and f_1 , so that $A_0/D_0 = -1/2$ and $A_2/D_0 = 5/2$. This behavior is physically expected: as the particle density approaches that of the host fluid.

5.2 Close-range limit

In the near-contact regime, the particles are separated by distances comparable to their diameter, while the acoustic wavelength remains much larger than the interparticle spacing. Accordingly, we consider $ka \lesssim kd \ll 1$. In this limit, the oscillatory factors in Eq. (30a) may be expanded for small kd , and the term proportional to $(kd)^2 \cos(kd)$ becomes subleading. As a consequence, the interaction force is governed at leading order by the ratio A_0/D_0 , yielding

$$\mathbf{F}^{\text{int}}|_{kd \ll 1} = 12\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \frac{A_0}{D_0} \mathbf{e}_x + O[(kd)^2]. \quad (33)$$

This expression shows that the close-range interaction retains the characteristic d^{-4} dependence of dipole-mediated nodal-plane forces, but with a prefactor renormalized by finite-separation multiple scattering through the ratio A_0/D_0 .

Using Eq. (31), we further obtain, for particles with densities close to that of the surrounding fluid,

$$\begin{aligned} \mathbf{F}^{\text{int}}|_{kd \ll 1, f_1 \ll 1} = & \\ -6\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 & \left[1 + \frac{1}{154} \left(77 \left(\frac{a}{d}\right)^3 + 990 \left(\frac{a}{d}\right)^7 + 3675 \left(\frac{a}{d}\right)^{11} \right) f_1 \right] \mathbf{e}_x + O(f_1^4). \end{aligned} \quad (34)$$

The leading term is therefore purely attractive and scales as $(a/d)^4$, whereas the bracketed correction gives the first finite-density-contrast contribution through higher powers of a/d . Taking the additional limit $f_1 \rightarrow 0$ in Eq. (34), we recover the joint asymptotic result

$$\mathbf{F}^{\text{int}}|_{kd \ll 1, f_1 \rightarrow 0} = -6\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \mathbf{e}_x. \quad (35)$$

Thus, in the combined close-range and weak-density-contrast limit, the interaction force reduces to a purely attractive dipolar law proportional to d^{-4} .

5.3 Long-range limit

For large interparticle separations, characterized by $a/d \ll 1$, the finite-separation corrections in Eq. (30a) become negligible at leading order, and the interaction force

reduces to

$$\begin{aligned} \mathbf{F}^{\text{int}}|_{a/d \ll 1} &= 4\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \left[-\frac{3}{2} [\cos(kd) + kd \sin(kd)] + \frac{1}{2} (kd)^2 \cos(kd) \right] \mathbf{e}_x \\ &+ \mathcal{O}\left[\left(\frac{a}{d}\right)^7\right]. \end{aligned} \quad (36)$$

This large-separation limit has the same leading-order functional form as Eq. (32), obtained independently in the weak-density-contrast regime. The coincidence is physically natural: in both cases the rescattering corrections become subdominant, either because dipolar coupling is weak or because the particles are geometrically far apart. Thus, as $a/d \rightarrow 0$, the amplitude ratios A_0/D_0 and A_2/D_0 approach constant values, and repeated mutual rescattering no longer affects the leading interaction force.

If, in addition, $kd \gg 1$, then the term proportional to $(kd)^2 \cos(kd)$ dominates in Eq. (36), and the interaction force further simplifies to

$$\mathbf{F}^{\text{int}}|_{a/d \ll 1, kd \gg 1} = 2\pi a^2 f_1^2 (ka)^2 E_0 \left(\frac{a}{d}\right)^2 \cos(kd) \mathbf{e}_x. \quad (37)$$

Hence, in the far-field regime $a/d \ll 1$ and $kd \gg 1$, the interaction force no longer follows the d^{-4} law of the close-range regime, but instead decays as d^{-2} , while retaining the oscillatory dependence on the acoustic phase through $\cos(kd)$.

6 Comparison with previous works

6.1 Earlier closed-form analytical results

The analytical result derived in Eq. (30a) may be compared with two earlier theoretical results that allow a direct formula-to-formula assessment in the present nodal-plane configuration. The first is the Rayleigh pair-potential theory of Silva & Bruus [18], which provides the standard analytical expression for the interaction force between two small particles trapped in a standing-wave nodal plane. The second is the earlier result for two particles in a nodal plane presented by Weiser *et al.* [15]. For historical accuracy, we note that the rigid-sphere interaction force used in that work is attributed by those authors to the earlier analysis of Nyborg & Gershoy (NG). Since that original source has not been examined directly in the present study, the comparison below is restricted to the formula as reproduced in [15].

6.1.1 Silva-Bruus interaction force

The expression of the acoustic interaction force given in Eq. (30a) can be compared directly with the earlier result derived by Silva & Bruus [18]. For two identical particles trapped in the nodal plane, their interaction force is

$$\mathbf{F}_{\text{SB}}^{\text{int}} = 4\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \left[-\frac{3}{2} [\cos(kd) + kd \sin(kd)] + \frac{1}{2} (kd)^2 \cos(kd) \right] \mathbf{e}_x. \quad (38)$$

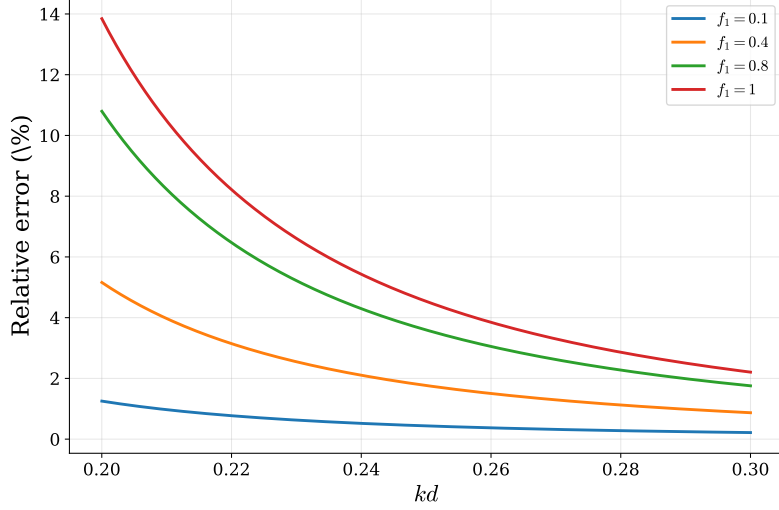


Fig. 2 Relative error between the present interaction-force model and the Silva–Bruus prediction as a function of kd , computed for $ka = 0.1$ and for four values of the dipole scattering factor, $f_1 = 0.1, 0.4, 0.8$, and 1 . The error increases with f_1 and decreases monotonically with kd over the close-range interval $2ka < kd < 0.3$.

Its connection with Eq. (30a) is immediate: the Silva–Bruus result is recovered from the present expression by setting $D_0 = 1$, $A_0 = -1/2$, and $A_2 = 5/2$. In the present theory, by contrast, these coefficients are replaced by the nontrivial ratios A_0/D_0 and A_2/D_0 , which depend explicitly on the separation ratio a/d . This is the mathematical signature of the finite-separation corrections introduced by repeated mutual rescattering.

The acoustic interaction force in (38) is recovered asymptotically in two distinct limits: the weak-density-contrast regime ($f_1 \ll 1$) and the large-separation regime ($a/d \ll 1$). This behavior is physically expected. In the former case, the dipolar coupling becomes weak as the particle density approaches that of the surrounding fluid; in the latter, the particles become sufficiently far apart that mutual rescattering is geometrically weak. In both limits, the interaction force reduces to the simpler Rayleigh pair-interaction force $\mathbf{F}_{\text{SB}}^{\text{int}}$.

Figure 2 shows the relative error between the present interaction-force model and $\mathbf{F}_{\text{SB}}^{\text{int}}$ for $ka = 0.1$. The error remains modest throughout the close-range interval $2ka < kd < 0.3$, but increases systematically with the dipole scattering factor f_1 . In all cases, the error is largest in the near-contact limit $kd \rightarrow 2ka$ and then decreases monotonically as kd increases. Quantitatively, the maximum error rises from about 1.25% for $f_1 = 0.1$ to 5.16%, 10.80%, and 13.84% for $f_1 = 0.4, 0.8$, and 1 , respectively. For $ka = 0.1$ the error remains below about 14% over the whole interval considered. Thus, in the small-particle regime the Silva–Bruus model [18] remains a reasonably accurate approximation, especially away from the immediate near-contact limit, whereas the present theory quantifies the finite-separation correction that becomes increasingly relevant as f_1 grows and d approaches $2a$.

6.1.2 Nyborg-Gershoy interaction force

For two rigid particles in a nodal plane, the NG interaction force exerted, say, on particle 1 by particle 2 can be expressed by [15]

$$\mathbf{F}_{\text{NG}}^{\text{int}} = -\frac{2\pi a^2}{3} \frac{(\rho_1 - \rho_0)^2}{\rho_0} \left(\frac{a}{d}\right)^4 |v(\mathbf{r}_{0,1})|^2 \mathbf{e}_x. \quad (39)$$

Using $E_0 = \rho_0 |v|^2/4$ and $f_1 = 2(\tilde{\rho} - 1)/(2\tilde{\rho} + 1)$, this expression becomes

$$\mathbf{F}_{\text{NG}}^{\text{int}} = -\frac{(2\tilde{\rho} + 1)^2}{9} 6\pi a^2 f_1^2 E_0 \left(\frac{a}{d}\right)^4 \mathbf{e}_x. \quad (40)$$

Hence, in the weak-density-contrast limit $\tilde{\rho} \approx 1$, or equivalently $f_1 \approx 0$, the NG interaction force becomes identical to $\mathbf{F}^{\text{int}}|_{kd \ll 1, f_1 \rightarrow 0}$ given by Eq. (35). Thus, both theories recover the same attractive nodal-plane dipolar law proportional to $(a/d)^4$. However, $\mathbf{F}_{\text{NG}}^{\text{int}}$ does not reproduce the oscillatory far-field behavior and the associated $(a/d)^2$ scaling predicted by Eq. (37). At present, no stronger conclusion is warranted, because the original derivation of Nyborg & Gershoy has not been examined directly here. Accordingly, the present comparison should be interpreted only as a comparison with the form reported in [15], not as a definitive assessment of the original NG theory.

6.2 Numerical and experimental data

6.2.1 Mohapatra, Sepehrirahnama *et al.* datasets

We next compare the close-range prediction in Eq. (34) with the numerical and experimental results reported in Refs. [21, 25, 29] for the acoustic interaction force between two equal polystyrene microparticles in water at the pressure nodal plane. In those works, the comparison is presented in terms of the normalized radial interaction force $\tilde{F}^{\text{int}} = F^{\text{int}}/(E_0 k^4 a^6)$, plotted as a function of kd in the close-range regime. The dataset comprises three experimental series, corresponding to particle diameters $2a = 7.81 \mu\text{m}$, $9.9 \mu\text{m}$, and $12.32 \mu\text{m}$, here denoted by EXP 1, EXP 2, and EXP 3, respectively, together with a numerical reference curve obtained from a fifth-order partial-wave expansion model, here denoted by NUM 5. For the theoretical benchmark reported in Ref. [21], the adopted parameters are $ka = 0.0314$, $\rho_p/\rho_f = 1.050$, and $c_p/c_f = 1.567$, which for polystyrene in water correspond to the dipole scattering factor $f_1 = 0.0323$. Under these conditions, the interaction force lies in the weak-density-contrast regime considered in Eq. (34). We therefore compare our analytical prediction, denoted by THEORY, with NUM 5 and EXP 1–EXP 3. For completeness, we also include the Silva–Bruus interaction force in Eq. (38), denoted by THEORY-SB.

To place all datasets on the same footing, we introduce the logarithmic variables $X = \log_{10}(kd)$ and $Y = \log_{10}|\tilde{F}^{\text{int}}|$. In terms of these variables, the experimental, numerical, and analytical trends are represented by the linear law $Y = C_1 X + C_2$. The coefficients (C_1, C_2) reported in Ref. [29] are EXP 1,2,3: $(-3.957, -1.246)$,

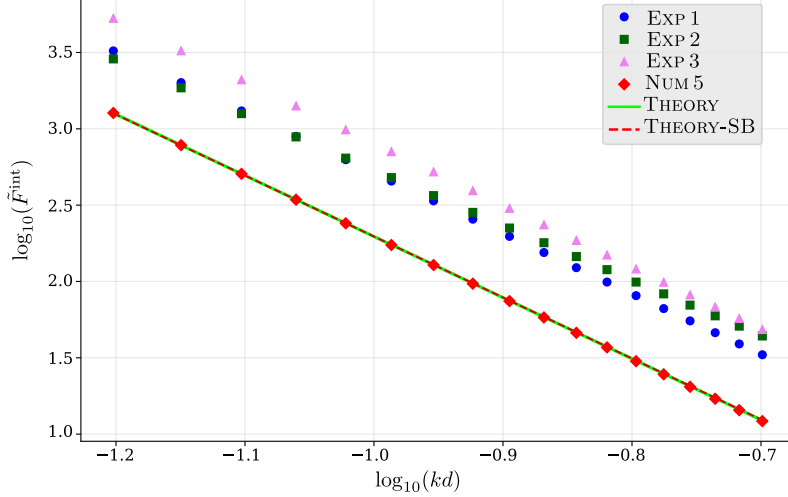


Fig. 3 Comparison of the normalized interaction force in the logarithmic variables $X = \log_{10}(kr)$ and $Y = \log_{10} \left[F_{1,x}^{\text{int}} / (E_0 k^4 a^6) \right]$. The numerical model NUM5 is shown by red diamonds, while the experimental datasets EXP 1, EXP 2, and EXP 3 are shown by blue circles, dark-green squares, and violet triangles, respectively. Models THEORY and THEORY-SB are nearly indistinguishable from NUM 5, whereas the experimental trends lie systematically above them.

$(-3.608, -0.879)$, $(-4.053, -1.148)$, and NUM 5: $(-4.013, -1.720)$. Moreover, least-squares fitting over the interval $2ka \leq kd \leq 0.2$ gives THEORY: $(-4.0020, -1.7078)$ and THEORY-SB: $(-3.9948, -1.7002)$.

Figure 3 shows that the present analytical model THEORY and the Silva–Bruus prediction THEORY-SB both follow an almost identical d^{-4} -type trend over the whole comparison range, with THEORY lying systematically but only slightly above THEORY-SB. The numerical reference NUM 5 is likewise very close to these two theoretical curves, with mean relative errors of about 0.55% between THEORY and NUM 5 and 0.84% between THEORY-SB and NUM 5. This indicates that all three ideal-fluid descriptions capture essentially the same close-range scaling. By contrast, the experimental datasets EXP 1–EXP 3 remain systematically above the theoretical curves throughout the plotted interval, although they also exhibit an approximately linear behavior. Quantitatively, the mean relative errors of THEORY with respect to EXP 1, EXP 2, and EXP 3 are about 62.1%, 66.1%, and 75.2%, respectively, while the corresponding values for THEORY-SB are 62.0%, 66.0%, and 75.2%.

Therefore, Fig. 3 indicates that, within the weak-density-contrast regime, the present model, the Silva–Bruus model [18], and the 5th-order partial-wave numerical calculation [21] become nearly indistinguishable at the level of the normalized close-range force. In this context, the substantial gap (50% difference) previously noted between the Silva–Bruus curve and later partial-wave numerical results [21] is due to a normalization issue rather than as evidence of a large physical discrepancy. In particular, the standing-wave energy scale in Ref. [18] is defined with a prefactor 1/2, whereas Refs. [21, 25, 29] use the acoustic energy density with prefactor 1/4. This

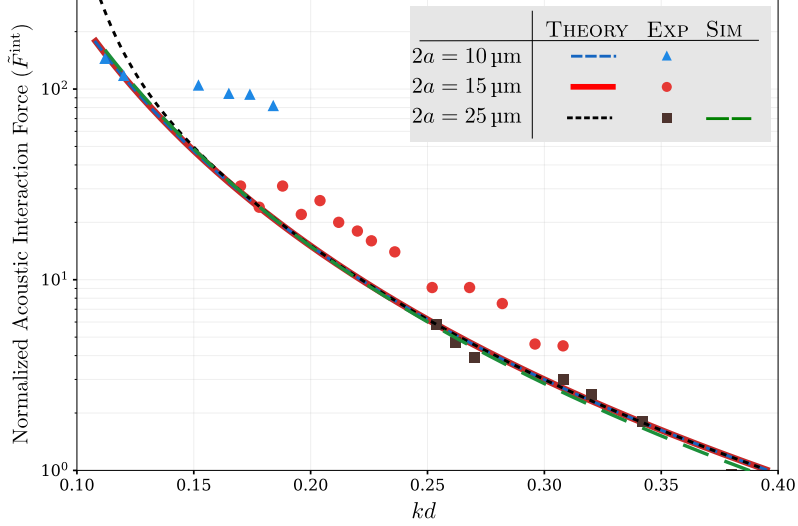


Fig. 4 Comparison between the present theory and the numerical and experimental results extracted from [27] for equal-sized polystyrene particles in water at the pressure nodal plane. The theoretical curves were evaluated from Eqs. (30a) and (31) using $f_1 = 0.0356$ and $ka = 0.0427$, 0.0640 , and 0.1067 . The dashed green curve represents the numerical simulation for the $25 \mu\text{m}$ particles, while the symbols denote the experimental datasets for $10 \mu\text{m}$, $15 \mu\text{m}$, and $25 \mu\text{m}$ particles.

factor-of-two difference shifts the normalized force by 50%, which is of the same order as the discrepancy emphasized in Ref. [21].

We finally remark that all ideal-fluid predictions remain substantially below the experimentally measured interaction force. This systematic deviation is consistent with the interpretation advanced in Ref. [22], namely that the experiments involve additional physical mechanisms not captured by ideal fluid approximations, in particular viscous effects, acoustic streaming, and wall-mediated hydrodynamic interactions, all of which may enhance the effective close-range attraction between the particles.

6.3 Hoque & Sen datasets

We next compare the present theory with the numerical and experimental results reported by Hoque & Sen [27] for equal-sized polystyrene particles in water located at the pressure nodal plane. From the material and acoustic parameters given in Ref. [27], we obtain the dipole scattering factor $f_1 = 0.0356$, while the size parameters $ka = 0.0427$, 0.0640 , and 0.1067 for $2a = 10 \mu\text{m}$, $15 \mu\text{m}$, and $25 \mu\text{m}$, respectively. Using these parameters in the acoustic interaction force given by Eqs. (30a) and (31), we evaluate the corresponding size-dependent interaction forces.

Figure 4 shows that the agreement between the present theory and the results from [27] improves systematically with increasing particle size. For the $25 \mu\text{m}$ particles, the theoretical curve lies very close to both the simulation and the experimental data over the whole plotted range, yielding a mean relative error of 9.75% with respect to the experimental points and 5.31% with respect to the simulation curve. By contrast, the discrepancies are significantly larger for the smaller particles, with mean relative

errors of 35.05% for 15 μm and 46.13% for 10 μm . Overall, the figure suggests that the present ideal-fluid theory becomes more accurate as the particle size increases. Conversely, the stronger deviations observed for the smaller particles indicate that effects not included in the present model, such as viscosity, acoustic streaming, and hydrodynamic interactions, become comparatively more important in that regime.

7 Summary & conclusions

We have derived a closed-form expression for the acoustic interaction force between two identical compressible particles trapped in the nodal plane of a standing-wave field. The analysis was carried out in the Rayleigh regime for an ideal fluid, so that the role of acoustic multiple scattering could be isolated. By exploiting the symmetry of the two-particle nodal-plane configuration, the coupled scattering problem was reduced to a tractable multipole system. At truncation order $N = 5$, this reduction leaves only six independent scattering coefficients, from which the interaction force was obtained explicitly in the form of Eq. (30a).

The asymptotic analysis showed that the present theory consistently reproduces the known limiting behaviors. In the close-range regime $ka \lesssim kd \ll 1$, the interaction retains the characteristic d^{-4} dependence, but with a prefactor renormalized by finite-separation multiple scattering. In the combined limit $kd \ll 1$ and $f_1 \rightarrow 0$, the force becomes the purely attractive dipolar law $\mathbf{F}^{\text{int}} = -6\pi a^2 f_1^2 E_0 (a/d)^4 \mathbf{e}_x$. By contrast, in the far-field regime $a/d \ll 1$ and $kd \gg 1$, the interaction no longer follows the close-range d^{-4} scaling, but instead decays as d^{-2} .

Comparison with previous analytical theories clarifies the role of the present result. The Silva–Bruus expression is recovered exactly when the finite-separation ratios take the constant values $D_0 = 1$, $A_0 = -1/2$, and $A_2 = 5/2$, and it is also recovered asymptotically in the weak-density-contrast and large-separation limits. For $ka = 0.1$, the relative error between the present model and the Silva–Bruus prediction remains below about 14% over the interval $2ka < kd < 0.3$, with maximum deviations of approximately 1.25%, 5.16%, 10.80%, and 13.84% for $f_1 = 0.1, 0.4, 0.8$, and 1, respectively. These values show that the Silva–Bruus theory [18] remains a good approximation, but that finite-separation rescattering corrections become increasingly relevant as the particles approach near contact and as the dipole contrast grows.

Comparison with numerical and experimental data further helps delimit the validity of the present ideal-fluid theory. For the close-range polystyrene-particle data reported in Refs. [21, 25, 29], corresponding to the weak-density-contrast regime $f_1 = 0.0323$, the present analytical model is nearly indistinguishable from both the Silva–Bruus theory and the fifth-order partial-wave numerical solution. The mean relative error between the present theory and the numerical reference is about 0.55%, while the corresponding error for the Silva–Bruus model is about 0.84%. By contrast, the error with respect to the experimental datasets can be as large 75% for the present theory. This indicates that the discrepancy with experiment is much larger and is unlikely to be explained by rescattering corrections alone. In this context, the present analysis also suggests that part of the discrepancy previously emphasized between the

Silva–Bruus curve and higher-order partial-wave numerical results [21] is due to an energy normalization issue rather than to a major physical incompleteness.

A second comparison, based on the data of Hoque & Sen [27], shows that the agreement between the present theory and the reported results improves systematically with increasing particle size. For particles of diameter 25 μm , the mean relative error is about 9.75% with respect to the experimental data and about 5.31% with respect to the numerical simulation. For smaller particles, however, the disagreement can be as large as 46% for 10 μm . This trend suggests that the present ideal-fluid multiple-scattering theory captures the dominant interaction mechanism more accurately for larger particles, while smaller particles are comparatively more sensitive to physical effects not included here, such as viscosity, acoustic streaming, and other near-field hydrodynamic interactions.

Taken together, the present theory provides a closed-form expression for the acoustic interaction force between two identical particles trapped in a nodal plane, including finite-separation multiple-scattering effects. This yields a compact description of pairwise interactions beyond the Silva–Bruus pair-potential model. It also suggests that, in dilute particle assemblies, the Silva–Bruus model should remain a useful approximation, whereas in denser nodal-plane structures genuinely many-body rescattering effects are expected to become relevant. The present study therefore provides a useful analytical benchmark for interpreting and modeling acoustic self-organization in standing-wave acoustofluidic systems.

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Data Availability. The datasets generated during this study are available from the corresponding author on reasonable request.

Disclosures. The author declares that there are no financial interests, or other potential conflicts of interest that could have influenced this research. ChatGPT was used as a support tool mainly for language polishing, text organization, and presentation. It also assisted with consistency checks, asymptotic discussion, comparisons with previous theories and published data, figure wording, captions, and general manuscript framing. The scientific conception, originality, problem selection, research direction, interpretation of the literature, mathematical derivations, physical conclusions, and final decisions about the manuscript are the sole responsibility of the author.

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Competing Interests. The author declares no competing interests.

Appendix A Single-particle scaled scattering coefficients

We shall derive the asymptotic form of first three odd single-particle scattering coefficient $s_n = s_n^{(i)}$ ($n = 1, 3, 5$). In the long-wavelength regime, $ka \ll 1$, we can use the small-argument expansions of the spherical Bessel and Hankel functions [32] in Eq. (10), and express the results in terms of the dipole scattering factor f_1 . Accordingly, we have

$$s_1 = i \frac{f_1}{6} (ka)^3 + O[(ka)^5], \quad (\text{A1a})$$

$$s_3 = \frac{i}{350} \frac{f_1}{7 - f_1} (ka)^7 + O[(ka)^9], \quad (\text{A1b})$$

$$s_5 = \frac{i}{1309770} \frac{f_1}{11 - 2f_1} (ka)^{11} + O[(ka)^{13}]. \quad (\text{A1c})$$

Appendix B Symmetry properties in two-particle configuration

We proceed to analyze how symmetry spatial properties of the two-particle problem can simplify the multiscattering linear system. First, the external standing wave $\tilde{p}^{\text{ext}} = \sin(kz)$ is independent of x and y , and is odd under the reflection $z \rightarrow -z$. Since both particle centers lie in the plane $z = 0$, the two-particle geometry is invariant under this reflection. Therefore, the total first-order pressure field must also be odd with respect to z . Using the parity property $Y_n^m(\pi - \theta, \phi) = (-1)^n Y_n^m(\theta, \phi)$, it follows that only odd multipole orders can be excited, so that

$$s_{nm}^{(i)} = 0, \quad n \text{ even.} \quad (\text{B2})$$

Second, the problem is invariant under reflection with respect to the xz plane, i.e. $y \rightarrow -y$. Under this transformation, $Y_n^{-m}(\theta, \phi) = (-1)^m Y_n^m(\theta, -\phi)$, and hence the scattering coefficients satisfy

$$s_{n,-m}^{(i)} = (-1)^m s_{nm}^{(i)}. \quad (\text{B3})$$

Thus, coefficients with opposite azimuthal order are not independent. Third, because the particles are identical and placed symmetrically with respect to the origin, the configuration is invariant under the reflection $x \rightarrow -x$, which exchanges particles 1 and 2. This symmetry implies

$$s_{nm}^{(1)} = s_{n,-m}^{(2)}. \quad (\text{B4})$$

Combining Eqs. (B3) and (B4), we obtain

$$s_{nm}^{(2)} = (-1)^m s_{nm}^{(1)}. \quad (\text{B5})$$

Finally, since the interparticle vector is parallel to the x axis, the separation coefficient $S_{n'n}^{m'm}$ preserves the parity of $n + m$. In the present standing-wave configuration, the external forcing at the nodal plane has only $m = 0$ and odd n , so that only channels with odd $n + m$ can be generated. Because only odd n survive by Eq. (B2), it follows that only even values of m are allowed, namely

$$s_{nm}^{(i)} = 0, \quad m \text{ odd}. \quad (\text{B6})$$

Under the symmetry constraints of the present two-particle problem

$$s_{n,-m}^{(i)} = s_{nm}^{(i)}, \quad s_{nm}^{(2)} = s_{nm}^{(1)}, \quad n \text{ odd}, \quad m \text{ even}, \quad (\text{B7})$$

the independent unknowns may be taken from a single particle only, with nonnegative even values of m . For a fixed odd multipole order n , the admissible values are $m = 0, 2, 4, \dots, n - 1$, so the number of independent coefficients at order n is $(n + 1)/2$. Therefore, if the multipole expansion is truncated at order N , the total number of independent unknowns is

$$\mathcal{N}(N) = \sum_{\substack{n=1 \\ n \text{ odd}}}^N \frac{n+1}{2} = \begin{cases} \frac{K(K+1)}{2}, & N = 2K, \\ \frac{(K+1)(K+2)}{2}, & N = 2K+1. \end{cases} \quad (\text{B8})$$

Appendix C Linear scattering coefficients system

For two-particle configuration in a nodal plane with $\mathbf{r}_{12} = -d\mathbf{e}_x$, we define the following matrices

$$\mathbf{t} = \begin{bmatrix} s_{10} \\ s_{30} \\ s_{32} \\ s_{50} \\ s_{52} \\ s_{54} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} \sqrt{12\pi} \\ -\sqrt{28\pi} \\ 0 \\ \sqrt{44\pi} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{D} = \text{diag}(s_1, s_3, s_3, s_5, s_5, s_5). \quad (\text{C9})$$

So the rescattering linear system for the two-particle nodal-plane problem given in (26) is

$$\mathbf{t} = \mathbf{D}(\mathbf{a} + \mathbf{T}\mathbf{t}) \quad \text{or} \quad \mathbf{t} = \mathbf{M}^{-1}\mathbf{D}\mathbf{a}, \quad (\text{C10})$$

where $\mathbf{M} = \mathbf{I} - \mathbf{T}'$ and the scaled translation matrix is $\mathbf{T}' = \mathbf{D}\mathbf{T}$. To construct the $\mathbf{T}'$ matrix, we assign row and column labels $\alpha, \beta \in \{1, \dots, 6\}$. Also, we define the index

transformation as

$$(1, 0) \mapsto 1, (3, 0) \mapsto 2, (3, 2) \mapsto 3, (5, 0) \mapsto 4, (5, 2) \mapsto 5, (5, 4) \mapsto 6,$$

so that, if $(n, m) \mapsto \alpha$ and $(n', m') \mapsto \beta$, then the matrix entry $T'_{\alpha\beta}$ is defined by

$$T'_{\alpha\beta} = D_\alpha \left[S_{n'n}^{m'm}(-\mathbf{d}) + (1 - \delta_{m',0}) S_{n'n}^{-m',m}(-\mathbf{d}) \right], \quad (\text{C11})$$

where $D_\alpha = (\mathbf{D})_{\alpha\alpha}$.

Using the explicit re-expansion formula for the separation coefficients in Eq. (13), we see that each matrix entry $T'_{\alpha\beta}$ becomes a finite linear combination of spherical Hankel functions of first kind. Therefore,

$$\mathbf{T}'(-\mathbf{d}) = \sum_{q=0}^{10} \mathbf{D} \mathbf{C}_q(-\mathbf{d}) h_q^{(1)}(kd), \quad (\text{C12})$$

where the constant matrices $\mathbf{C}_q$ collect the angular and Gaunt-coupling factors,

$$\begin{aligned} (\mathbf{C}_q)_{\alpha\beta} = 4\pi i^{n-n'+q} & \left[Y_q^{m-m'} [\Omega(-\mathbf{d})] \mathcal{G}(n', m'; q, m-m'; n, m) \right. \\ & \left. + (1 - \delta_{m',0}) Y_q^{m+m'} [\Omega(-\mathbf{d})] \mathcal{G}(n', -m'; q, m+m'; n, m) \right], \end{aligned} \quad (\text{C13})$$

for the values of q allowed by the angular-momentum selection rules, and $(\mathbf{C}_q)_{\alpha\beta} = 0$ otherwise. Note also that $\Omega(-\mathbf{d}) = (\pi/2, \pi)$. Only even values of q contribute to the Hankel expansion of the scaled translation matrix. Hence, the odd-order matrices vanish, $\mathbf{C}_1 = \mathbf{C}_3 = \mathbf{C}_5 = \mathbf{C}_7 = \mathbf{C}_9 = \mathbf{0}$, and Eq. (C12) becomes

$$\mathbf{T}'(-\mathbf{d}) = \sum_{\ell=0}^5 \mathbf{D} \mathbf{C}_{2\ell}(-\mathbf{d}) h_{2\ell}^{(1)}(kd), \quad (\text{C14})$$

The $\mathbf{M}$ matrix is computed algebraically using MATHEMATICA Software (ver. 13, Wolfram Research Inc., USA). In so doing, we use the asymptotic formula $h_n^{(1)}(x) \sim -i \frac{(2n-1)!!}{x^{n+1}}$, $x \rightarrow 0$, and introduce the dimensionless parameters $\chi = a/d$ and $\epsilon = ka$. For clarity, we display $\mathbf{M}$ in the block form

$$\mathbf{M} = \begin{pmatrix} M_{11} & \mathbf{M}_{12} & \mathbf{M}_{13} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} \\ \mathbf{M}_{31} & \mathbf{M}_{32} & \mathbf{M}_{33} \end{pmatrix}, \quad (\text{C15})$$

where M_{11} is a scalar block, $\mathbf{M}_{22}$ is a 2×2 block, and $\mathbf{M}_{33}$ is a 3×3 block. Explicitly, the matrix entries are

$$M_{11} = 1 - \frac{f_1 \chi^3}{2} - \frac{1}{6} f_1 \epsilon^2 \chi,$$

$$\mathbf{M}_{12} = (M_{12} \ M_{13}), \quad \mathbf{M}_{13} = (M_{14} \ M_{15} \ M_{16}),$$

with

$$\begin{aligned} M_{12} &= \frac{15f_1\sqrt{21}\chi^5}{4\epsilon^2} + \frac{3}{4}f_1\sqrt{\frac{3}{7}}\chi^3, \quad M_{13} = -\frac{15\sqrt{\frac{35}{2}}f_1\chi^5}{2\epsilon^2} - \frac{3}{2}f_1\sqrt{\frac{5}{14}}\chi^3, \\ M_{14} &= -\frac{4725\sqrt{33}f_1\chi^7}{16\epsilon^4} - \frac{525\sqrt{\frac{3}{11}}f_1\chi^5}{16\epsilon^2}, \quad M_{15} = \frac{945f_1\sqrt{\frac{385}{2}}\chi^7}{4\epsilon^4} + \frac{105f_1\sqrt{\frac{35}{22}}\chi^5}{4\epsilon^2}, \\ M_{16} &= -\frac{945\sqrt{\frac{1155}{2}}f_1\chi^7}{8\epsilon^4} - \frac{105\sqrt{\frac{105}{22}}f_1\chi^5}{8\epsilon^2}. \end{aligned}$$

$$\mathbf{M}_{21} = \begin{pmatrix} M_{21} \\ M_{31} \end{pmatrix}, \quad \mathbf{M}_{22} = \begin{pmatrix} M_{22} & M_{23} \\ M_{32} & M_{33} \end{pmatrix}, \quad \mathbf{M}_{23} = \begin{pmatrix} \mathbf{m}_{23}^{(1)} \\ \mathbf{m}_{23}^{(2)} \end{pmatrix},$$

with

$$M_{21} = \frac{9f_1\epsilon^2\sqrt{\frac{3}{7}}\chi^5}{20(7-f_1)} + \frac{9f_1\epsilon^4\sqrt{\frac{3}{7}}\chi^3}{700(7-f_1)}, \quad M_{31} = -\frac{9f_1\epsilon^2\chi^5}{4\sqrt{70}(7-f_1)} - \frac{9f_1\epsilon^4\chi^3}{140\sqrt{70}(7-f_1)},$$

$$\begin{aligned} M_{22} &= 1 - \frac{225f_1\chi^7}{8(7-f_1)} - \frac{81f_1\epsilon^2\chi^5}{440(7-f_1)} - \frac{f_1\epsilon^4\chi^3}{175(7-f_1)} - \frac{f_1\epsilon^6\chi}{350(7-f_1)}, \\ M_{23} &= \frac{63f_1\sqrt{\frac{15}{2}}\chi^7}{4(7-f_1)} - \frac{9\sqrt{\frac{3}{10}}f_1\epsilon^2\chi^5}{44(7-f_1)} - \frac{\sqrt{\frac{3}{10}}f_1\epsilon^4\chi^3}{35(7-f_1)}, \\ M_{32} &= \frac{63f_1\sqrt{\frac{15}{2}}\chi^7}{8(7-f_1)} - \frac{9\sqrt{\frac{3}{10}}f_1\epsilon^2\chi^5}{88(7-f_1)} - \frac{\sqrt{\frac{3}{10}}f_1\epsilon^4\chi^3}{70(7-f_1)}, \\ M_{33} &= 1 - \frac{351f_1\chi^7}{8(7-f_1)} - \frac{63f_1\epsilon^2\chi^5}{440(7-f_1)} - \frac{f_1\epsilon^6\chi}{350(7-f_1)}. \end{aligned}$$

$$\mathbf{m}_{23}^{(1)} = (M_{24} \ M_{25} \ M_{26}), \quad \mathbf{m}_{23}^{(2)} = (M_{34} \ M_{35} \ M_{36}),$$

with

$$\begin{aligned} M_{24} &= \frac{19845f_1\sqrt{77}\chi^9}{32(7-f_1)\epsilon^2} + \frac{63f_1\sqrt{77}\chi^7}{32(7-f_1)} + \frac{81f_1\epsilon^2\chi^5}{52\sqrt{77}(7-f_1)} + \frac{f_1\epsilon^4\chi^3}{14\sqrt{77}(7-f_1)}, \\ M_{25} &= -\frac{8505\sqrt{\frac{165}{2}}f_1\chi^9}{8(7-f_1)\epsilon^2} - \frac{63\sqrt{\frac{33}{10}}f_1\chi^7}{8(7-f_1)} - \frac{9\sqrt{\frac{3}{110}}f_1\epsilon^2\chi^5}{26(7-f_1)} - \frac{\sqrt{\frac{3}{110}}f_1\epsilon^4\chi^3}{14(7-f_1)}, \\ M_{26} &= \frac{18711f_1\sqrt{\frac{55}{2}}\chi^9}{16(7-f_1)\epsilon^2} - \frac{567\sqrt{\frac{11}{10}}f_1\chi^7}{16(7-f_1)} - \frac{189f_1\epsilon^2\chi^5}{26\sqrt{110}(7-f_1)}, \end{aligned}$$

$$\begin{aligned}
M_{34} &= -\frac{5103\sqrt{\frac{1155}{2}}f_1\chi^9}{32(7-f_1)\epsilon^2} + \frac{63f_1\sqrt{\frac{231}{10}}\chi^7}{32(7-f_1)} + \frac{9f_1\epsilon^2\sqrt{\frac{15}{154}}\chi^5}{13(7-f_1)} - \frac{\sqrt{\frac{3}{770}}f_1\epsilon^4\chi^3}{28(7-f_1)}, \\
M_{35} &= \frac{37989f_1\sqrt{11}\chi^9}{16(7-f_1)\epsilon^2} - \frac{189\sqrt{11}f_1\chi^7}{80(7-f_1)} - \frac{45f_1\epsilon^2\chi^5}{52\sqrt{11}(7-f_1)} + \frac{3f_1\epsilon^4\chi^3}{140\sqrt{11}(7-f_1)}, \\
M_{36} &= -\frac{44793\sqrt{33}f_1\chi^9}{32(7-f_1)\epsilon^2} - \frac{567\sqrt{33}f_1\chi^7}{160(7-f_1)} - \frac{9\sqrt{\frac{3}{11}}f_1\epsilon^2\chi^5}{26(7-f_1)} - \frac{3\sqrt{\frac{3}{11}}f_1\epsilon^4\chi^3}{140(7-f_1)}.
\end{aligned}$$

$$\mathbf{M}_{31} = \begin{pmatrix} M_{41} \\ M_{51} \\ M_{61} \end{pmatrix}, \quad \mathbf{M}_{32} = \begin{pmatrix} \mathbf{m}_{32}^{(1)} \\ \mathbf{m}_{32}^{(2)} \\ \mathbf{m}_{32}^{(3)} \end{pmatrix}, \quad \mathbf{M}_{33} = \begin{pmatrix} \mathbf{m}_{33}^{(1)} \\ \mathbf{m}_{33}^{(2)} \\ \mathbf{m}_{33}^{(3)} \end{pmatrix},$$

with

$$\begin{aligned}
M_{41} &= -\frac{5f_1\epsilon^4\chi^7}{112\sqrt{33}(11-2f_1)} - \frac{5f_1\epsilon^6\chi^5}{11088\sqrt{33}(11-2f_1)}, \\
M_{51} &= \frac{f_1\epsilon^4\sqrt{\frac{5}{154}}\chi^7}{24(11-2f_1)} + \frac{f_1\epsilon^6\sqrt{\frac{5}{154}}\chi^5}{2376(11-2f_1)}, \quad M_{61} = -\frac{\sqrt{\frac{5}{462}}f_1\epsilon^4\chi^7}{16(11-2f_1)} - \frac{\sqrt{\frac{5}{462}}f_1\epsilon^6\chi^5}{1584(11-2f_1)}.
\end{aligned}$$

$$\mathbf{m}_{32}^{(1)} = (M_{42} \ M_{43}), \quad \mathbf{m}_{32}^{(2)} = (M_{52} \ M_{53}), \quad \mathbf{m}_{32}^{(3)} = (M_{62} \ M_{63}),$$

with

$$\begin{aligned}
M_{42} &= \frac{175f_1\epsilon^2\sqrt{\frac{7}{11}}\chi^9}{96(11-2f_1)} + \frac{5f_1\epsilon^4\sqrt{\frac{7}{11}}\chi^7}{864(11-2f_1)} + \frac{5f_1\epsilon^6\chi^5}{12012\sqrt{77}(11-2f_1)} \\
&\quad + \frac{5f_1\epsilon^8\chi^3}{261954\sqrt{77}(11-2f_1)}, \\
M_{43} &= -\frac{15\sqrt{\frac{105}{22}}f_1\epsilon^2\chi^9}{16(11-2f_1)} + \frac{f_1\epsilon^4\sqrt{\frac{35}{66}}\chi^7}{144(11-2f_1)} + \frac{5f_1\epsilon^6\sqrt{\frac{10}{231}}\chi^5}{9009(11-2f_1)} - \frac{\sqrt{\frac{5}{462}}f_1\epsilon^8\chi^3}{87318(11-2f_1)}, \\
M_{52} &= -\frac{25\sqrt{\frac{15}{22}}f_1\epsilon^2\chi^9}{16(11-2f_1)} - \frac{\sqrt{\frac{5}{66}}f_1\epsilon^4\chi^7}{144(11-2f_1)} - \frac{\sqrt{\frac{5}{66}}f_1\epsilon^6\chi^5}{36036(11-2f_1)} - \frac{\sqrt{\frac{5}{66}}f_1\epsilon^8\chi^3}{174636(11-2f_1)}, \\
M_{53} &= \frac{335f_1\epsilon^2\chi^9}{48\sqrt{11}(11-2f_1)} - \frac{f_1\epsilon^4\chi^7}{144\sqrt{11}(11-2f_1)} - \frac{25f_1\epsilon^6\chi^5}{108108\sqrt{11}(11-2f_1)} \\
&\quad + \frac{f_1\epsilon^8\chi^3}{174636\sqrt{11}(11-2f_1)}, \\
M_{62} &= \frac{5f_1\epsilon^2\sqrt{\frac{55}{2}}\chi^9}{32(11-2f_1)} - \frac{\sqrt{\frac{5}{22}}f_1\epsilon^4\chi^7}{96(11-2f_1)} - \frac{\sqrt{\frac{5}{22}}f_1\epsilon^6\chi^5}{5148(11-2f_1)}, \\
M_{63} &= -\frac{395f_1\epsilon^2\chi^9}{32\sqrt{33}(11-2f_1)} - \frac{f_1\epsilon^4\chi^7}{32\sqrt{33}(11-2f_1)} - \frac{5f_1\epsilon^6\chi^5}{18018\sqrt{33}(11-2f_1)}
\end{aligned}$$

$$-\frac{f_1 \epsilon^8 \chi^3}{58212\sqrt{33}(11-2f_1)}.$$

$$\mathbf{m}_{33}^{(1)} = (M_{44} \ M_{45} \ M_{46}),$$

with

$$\begin{aligned} M_{44} &= 1 - \frac{59535f_1\chi^{11}}{128(11-2f_1)} - \frac{6125f_1\epsilon^2\chi^9}{7296(11-2f_1)} - \frac{25f_1\epsilon^4\chi^7}{6426(11-2f_1)} - \frac{f_1\epsilon^6\chi^5}{24024(11-2f_1)} \\ &\quad - \frac{5f_1\epsilon^8\chi^3}{3405402(11-2f_1)} - \frac{f_1\epsilon^{10}\chi}{1309770(11-2f_1)}, \\ M_{45} &= \frac{3465f_1\sqrt{\frac{105}{2}}\chi^{11}}{32(11-2f_1)} + \frac{25f_1\epsilon^2\sqrt{\frac{105}{2}}\chi^9}{608(11-2f_1)} - \frac{5\sqrt{\frac{5}{42}}f_1\epsilon^4\chi^7}{612(11-2f_1)} - \frac{\sqrt{\frac{5}{42}}f_1\epsilon^6\chi^5}{5148(11-2f_1)} \\ &\quad - \frac{\sqrt{\frac{5}{42}}f_1\epsilon^8\chi^3}{81081(11-2f_1)}, \\ M_{46} &= -\frac{6435\sqrt{\frac{35}{2}}f_1\chi^{11}}{64(11-2f_1)} + \frac{605f_1\epsilon^2\sqrt{\frac{35}{2}}\chi^9}{1216(11-2f_1)} + \frac{f_1\epsilon^4\sqrt{\frac{5}{14}}\chi^7}{102(11-2f_1)} - \frac{\sqrt{\frac{5}{14}}f_1\epsilon^6\chi^5}{5148(11-2f_1)}. \end{aligned}$$

$$\mathbf{m}_{33}^{(2)} = (M_{54} \ M_{55} \ M_{56}),$$

with

$$\begin{aligned} M_{54} &= \frac{3465f_1\sqrt{\frac{105}{2}}\chi^{11}}{64(11-2f_1)} + \frac{25f_1\epsilon^2\sqrt{\frac{105}{2}}\chi^9}{1216(11-2f_1)} - \frac{5\sqrt{\frac{5}{42}}f_1\epsilon^4\chi^7}{1224(11-2f_1)} - \frac{\sqrt{\frac{5}{42}}f_1\epsilon^6\chi^5}{10296(11-2f_1)} \\ &\quad - \frac{\sqrt{\frac{5}{42}}f_1\epsilon^8\chi^3}{162162(11-2f_1)}, \\ M_{55} &= 1 - \frac{23175f_1\chi^{11}}{32(11-2f_1)} - \frac{245f_1\epsilon^2\chi^9}{1824(11-2f_1)} - \frac{f_1\epsilon^4\chi^7}{714(11-2f_1)} - \frac{2f_1\epsilon^6\chi^5}{27027(11-2f_1)} \\ &\quad - \frac{f_1\epsilon^8\chi^3}{1135134(11-2f_1)} - \frac{f_1\epsilon^{10}\chi}{1309770(11-2f_1)}, \\ M_{56} &= \frac{18975f_1\sqrt{3}\chi^{11}}{64(11-2f_1)} - \frac{2195f_1\epsilon^2\chi^9}{1216\sqrt{3}(11-2f_1)} - \frac{f_1\epsilon^4\chi^7}{136\sqrt{3}(11-2f_1)} + \frac{5f_1\epsilon^6\chi^5}{72072\sqrt{3}(11-2f_1)} \\ &\quad - \frac{f_1\epsilon^8\chi^3}{378378\sqrt{3}(11-2f_1)}. \end{aligned}$$

$$\mathbf{m}_{33}^{(3)} = (M_{64} \ M_{65} \ M_{66}),$$

with

$$M_{64} = -\frac{6435\sqrt{\frac{35}{2}}f_1\chi^{11}}{128(11-2f_1)} + \frac{605f_1\epsilon^2\sqrt{\frac{35}{2}}\chi^9}{2432(11-2f_1)} + \frac{f_1\epsilon^4\sqrt{\frac{5}{14}}\chi^7}{204(11-2f_1)} - \frac{\sqrt{\frac{5}{14}}f_1\epsilon^6\chi^5}{10296(11-2f_1)},$$

$$\begin{aligned}
M_{65} &= \frac{18975f_1\sqrt{3}\chi^{11}}{64(11-2f_1)} - \frac{2195f_1\epsilon^2\chi^9}{1216\sqrt{3}(11-2f_1)} - \frac{f_1\epsilon^4\chi^7}{136\sqrt{3}(11-2f_1)} + \frac{5f_1\epsilon^6\chi^5}{72072\sqrt{3}(11-2f_1)} \\
&\quad - \frac{f_1\epsilon^8\chi^3}{378378\sqrt{3}(11-2f_1)}, \\
M_{66} &= 1 - \frac{93525f_1\chi^{11}}{128(11-2f_1)} - \frac{13375f_1\epsilon^2\chi^9}{7296(11-2f_1)} - \frac{5f_1\epsilon^4\chi^7}{1071(11-2f_1)} + \frac{f_1\epsilon^6\chi^5}{24024(11-2f_1)} \\
&\quad + \frac{f_1\epsilon^8\chi^3}{1135134(11-2f_1)} - \frac{f_1\epsilon^{10}\chi}{1309770(11-2f_1)}.
\end{aligned}$$

Substituting $\mathbf{M}$ into (C10) along with (A1) and solving the obtained linear system, while retaining only the lowest order terms in ϵ , one finds

$$\begin{aligned}
s_{10} &= if_1\sqrt{\frac{\pi}{3}}\epsilon^3c_{10}, \quad s_{30} = if_1^2\sqrt{\frac{\pi}{7}}\chi^5\epsilon^5c_{30}, \quad s_{32} = if_1^2\sqrt{\frac{6\pi}{35}}\chi^5\epsilon^5c_{32}, \\
s_{50} &= if_1^2\sqrt{\frac{\pi}{11}}\chi^7\epsilon^7c_{50}, \quad s_{52} = if_1^2\sqrt{\frac{5\pi}{462}}\chi^7\epsilon^7c_{52}, \quad s_{54} = if_1^2\sqrt{\frac{5\pi}{154}}\chi^7\epsilon^7c_{54}. \quad (\text{C16})
\end{aligned}$$

with the dimensionless coefficients

$$\begin{aligned}
c_{10} &= -\frac{4}{D_c} \left(154 - 50f_1 + 4f_1^2 - 1485f_1\chi^7 + 270f_1^2\chi^7 - 22050f_1\chi^{11} + 3150f_1^2\chi^{11} \right. \\
&\quad \left. + 80325f_1^2\chi^{18} \right), \quad c_{30} = -\frac{18}{5} \frac{-11 + 2f_1 + 1050f_1\chi^{11}}{D_c}, \\
c_{32} &= 3 \frac{-11 + 2f_1 + 1050f_1\chi^{11}}{D_c}, \quad c_{50} = -\frac{5}{84} \frac{14 - 2f_1 + 117f_1\chi^7}{D_c}, \\
c_{52} &= \frac{1}{6} \frac{14 - 2f_1 + 117f_1\chi^7}{D_c}, \quad c_{54} = -\frac{1}{12} \frac{14 - 2f_1 + 117f_1\chi^7}{D_c}, \quad (\text{C17})
\end{aligned}$$

where the common denominator is

$$\begin{aligned}
D_c &= -616 + 200f_1 - 16f_1^2 + 308f_1\chi^3 - 100f_1^2\chi^3 + 8f_1^3\chi^3 + 5940f_1\chi^7 - 1080f_1^2\chi^7 \\
&\quad - 1782f_1^2\chi^{10} + 324f_1^3\chi^{10} + 88200f_1\chi^{11} - 12600f_1^2\chi^{11} - 40950f_1^2\chi^{14} \\
&\quad + 5850f_1^3\chi^{14} - 321300f_1^2\chi^{18} + 73575f_1^3\chi^{21}. \quad (\text{C18})
\end{aligned}$$

In the limit of large interparticle separation, $d \rightarrow \infty$, one has $\chi \rightarrow 0$, so that all multiple-scattering corrections vanish. In this regime,

$$s_{10} \xrightarrow{\chi \rightarrow 0} if_1\sqrt{\frac{\pi}{3}}\epsilon^3, \quad s_{30}, s_{32}, s_{50}, s_{52}, s_{54} \xrightarrow{\chi \rightarrow 0} 0. \quad (\text{C19})$$

Appendix D The $b_{21}^{(2 \rightarrow 1)}$ scattering coefficient

For the two-particle nodal-plane problem truncated at multipole order $N = 5$, the independent scattering coefficients that survive the symmetry reduction are $s_{10}, s_{30}, s_{32}, s_{50}, s_{52}, s_{54}$. Accordingly, the coefficient $b_{21}^{(2 \rightarrow 1)}$ in Eq. (26) becomes

$$\begin{aligned} b_{21}^{(2 \rightarrow 1)} = & S_{1,2}^{0,1}(-\mathbf{d}) s_{10} + S_{3,2}^{0,1}(-\mathbf{d}) s_{30} + \left[S_{3,2}^{2,1}(-\mathbf{d}) + S_{3,2}^{-2,1}(-\mathbf{d}) \right] s_{32} \\ & + S_{5,2}^{0,1}(-\mathbf{d}) s_{50} + \left[S_{5,2}^{2,1}(-\mathbf{d}) + S_{5,2}^{-2,1}(-\mathbf{d}) \right] s_{52} \\ & + \left[S_{5,2}^{4,1}(-\mathbf{d}) + S_{5,2}^{-4,1}(-\mathbf{d}) \right] s_{54}. \end{aligned} \quad (\text{D20})$$

To proceed with the separation coefficients calculation, we note that $r_{21} = d$ and $\Omega_{21} = (\frac{\pi}{2}, \pi)$. Using Eq. (13), with $n = 2$ and $m = 1$, the separation coefficients entering $b_{21}^{(2 \rightarrow 1)}$ are

$$\begin{aligned} S_{1,2}^{0,1}(-\mathbf{d}) &= \frac{3\sqrt{10}}{10} h_1 + \frac{3\sqrt{10}}{10} h_3, \\ S_{3,2}^{0,1}(-\mathbf{d}) &= \frac{3\sqrt{210}}{70} h_1 - \frac{\sqrt{210}}{60} h_3 - \frac{5\sqrt{210}}{84} h_5, \\ S_{3,2}^{2,1}(-\mathbf{d}) &= -\frac{3\sqrt{7}}{7} h_1 - \frac{\sqrt{7}}{4} h_3 + \frac{5\sqrt{7}}{28} h_5, \\ S_{3,2}^{-2,1}(-\mathbf{d}) &= \frac{5\sqrt{7}}{12} h_3 + \frac{5\sqrt{7}}{12} h_5, \\ S_{5,2}^{0,1}(-\mathbf{d}) &= -\frac{5\sqrt{330}}{132} h_3 + \frac{5\sqrt{330}}{624} h_5 + \frac{105\sqrt{330}}{2288} h_7, \\ S_{5,2}^{2,1}(-\mathbf{d}) &= \frac{5\sqrt{77}}{44} h_3 + \frac{5\sqrt{77}}{104} h_5 - \frac{75\sqrt{77}}{1144} h_7, \\ S_{5,2}^{-2,1}(-\mathbf{d}) &= \frac{5\sqrt{77}}{132} h_3 - \frac{25\sqrt{77}}{312} h_5 - \frac{135\sqrt{77}}{1144} h_7, \\ S_{5,2}^{4,1}(-\mathbf{d}) &= -\frac{5\sqrt{231}}{66} h_3 - \frac{35\sqrt{231}}{624} h_5 + \frac{45\sqrt{231}}{2288} h_7, \\ S_{5,2}^{-4,1}(-\mathbf{d}) &= \frac{15\sqrt{231}}{208} h_5 + \frac{15\sqrt{231}}{208} h_7. \end{aligned} \quad (\text{D21})$$

Hence, the combinations entering Eq. (D20) are

$$\begin{aligned} S_{3,2}^{2,1}(-\mathbf{d}) + S_{3,2}^{-2,1}(-\mathbf{d}) &= -\frac{3\sqrt{7}}{7} h_1 + \frac{\sqrt{7}}{6} h_3 + \frac{25\sqrt{7}}{42} h_5, \\ S_{5,2}^{2,1}(-\mathbf{d}) + S_{5,2}^{-2,1}(-\mathbf{d}) &= \frac{5\sqrt{77}}{33} h_3 - \frac{5\sqrt{77}}{156} h_5 - \frac{105\sqrt{77}}{572} h_7, \end{aligned} \quad (\text{D22})$$

$$S_{5,2}^{4,1}(-\mathbf{d}) + S_{5,2}^{-4,1}(-\mathbf{d}) = -\frac{5\sqrt{231}}{66} h_3 + \frac{5\sqrt{231}}{312} h_5 + \frac{105\sqrt{231}}{1144} h_7.$$

Equation (D20) is arranged as

$$b_{21}^{(2 \rightarrow 1)} = h_1(kd) B_1 + h_3(kd) B_3 + h_5(kd) B_5 + h_7(kd) B_7, \quad (\text{D23})$$

with

$$\begin{aligned} B_1 &= \frac{3}{\sqrt{10}} s_{10} + 3\sqrt{\frac{3}{70}} s_{30} - \frac{3}{\sqrt{7}} s_{32}, \\ B_3 &= \frac{3}{\sqrt{10}} s_{10} - \frac{1}{2}\sqrt{\frac{7}{30}} s_{30} + \frac{\sqrt{7}}{6} s_{32} - \frac{5\sqrt{330}}{132} s_{50} + \frac{5\sqrt{77}}{33} s_{52} - \frac{5\sqrt{231}}{66} s_{54}, \\ B_5 &= -\frac{5\sqrt{210}}{84} s_{30} + \frac{25\sqrt{7}}{42} s_{32} + \frac{5\sqrt{330}}{624} s_{50} - \frac{5\sqrt{77}}{156} s_{52} + \frac{5\sqrt{231}}{312} s_{54}, \\ B_7 &= \frac{105\sqrt{330}}{2288} s_{50} - \frac{105\sqrt{77}}{572} s_{52} + \frac{105\sqrt{231}}{1144} s_{54}. \end{aligned} \quad (\text{D24})$$

After substituting the reduced $N = 5$ scattering coefficients from Eq. (C) into (D24), all terms of (D23) can be reorganized into

$$b_{21}^{(2 \rightarrow 1)} = \frac{3}{\sqrt{210}} \frac{f_1 \chi^2}{D_b} \frac{e^{i\epsilon/\chi}}{\epsilon} [15\chi A_0 (\chi - i\epsilon) - A_2 \epsilon^2], \quad (\text{D25})$$

where

$$\begin{aligned} A_0 &= -308 + f_1 [100 - 8f_1 + 90(11 - 2f_1)\chi^7 - 5250(-7 + f_1)\chi^{11} - 33075f_1\chi^{18}], \\ A_2 &= 21 [-1540 + f_1(500 - 40f_1 + 162(11 - 2f_1)\chi^7 - 24250(-7 + f_1)\chi^{11} \\ &\quad + 20025f_1\chi^{18})], \\ D_b &= -616 + 4f_1 [50 + 77\chi^3 + 45\chi^7(33 + 490\chi^4)] + f_1^3\chi^3 [8 + 9\chi^7(36 + 650\chi^4 \\ &\quad + 8175\chi^{11})] - 2f_1^2 [8 + 50\chi^3 + 9\chi^7(60 + \chi^3(99 + 175\chi(4 + 13\chi^3 + 102\chi^7)))]]. \end{aligned} \quad (\text{D26})$$

In (D25), only the polynomial factor in brackets is retained through $O(\epsilon^2)$; the exponential factor $e^{i\epsilon/\chi}$ is left unexpanded.

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